



Muhammad Umer Sajid

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<https://scientist444.github.io/portfolio/> |

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ABOUT ME

Enthusiastic Electrical Engineering student with a strong academic record (**CGPA: 3.79**) and hands-on experience in embedded systems and hardware design. Proven ability to drive innovation through collaborative problem-solving and project management. Passionate about **Robotics and Artificial Intelligence**, with a commitment to developing technology that makes a tangible impact in the field.

EDUCATION AND TRAINING

26/09/2022 – CURRENT Islamabad, Pakistan

BACHELOR OF ELECTRICAL ENGINEERING NUST

- Current CGPA: 3.79 / 4.00
- Expected Graduation: June 2026

Website <https://nust.edu.pk/> | **Field of study** Electronics and automation , Electricity and energy | **Level in EQF** EQF level 6

WORK EXPERIENCE

ELECTRICAL ENGINEER – RISETECH – 01/02/2024 – Current – ISLAMABAD, PAKISTAN

Concurrently executing complex, production-grade hardware engineering responsibilities while maintaining full-time enrollment in a rigorous Bachelor of Electrical Engineering program.

- **High-Speed Design & Signal Integrity:** Engineered high-speed communication interfaces for medical devices, ensuring strict compliance with **EN 60601-1** standards and enhancing **Signal Integrity** by precisely calculating **90 Ω controlled impedance** differential pairs for USB 2.0.
- **Production-Grade Hardware Development:** Executed the end-to-end development of **4 production-ready systems**, managing the full lifecycle from complex multi-layer schematic capture in EasyEDA to manufacturing coordination with JLCPCB, reducing overall prototyping time.
- **Low-Level Firmware & Control Systems:** Architected and programmed robust firmware for **STM32 and Renesas** microcontrollers using STM32CubeIDE and e2 studio, implementing optimized **Control Systems** logic that improved system responsiveness for therapeutic devices.
- **Power Network Engineering:** Designed ESP32-based architectures integrating **BLE and Wi-Fi communication protocols**, while managing highly dynamic power domains scaling from **100 mA sleep states to 12 A DC loads** with verified stability.
- **Agile Project Management:** Successfully maintained a **100% project delivery rate** on strict industry deadlines while balancing a full-time academic load and a **3.79 CGPA**, demonstrating elite time management and professional reliability

EMBEDDED SYSTEM DESIGNER – RESEARCH INSTRUMENTS & MEASUREMENT SYSTEMS (RIMS) – 01/06/2025 – 01/08/2025 – ISLAMABAD, PAKISTAN

Operated within a high-stakes aerospace and defense R&D department, translating theoretical RF, embedded, and avionics concepts into practical, deployable engineering solutions.

- **Aerospace & Avionics Test Engineering:** Developed Avionics Line Replaceable Unit (LRU) and Portable Tester Automated Test Procedure (ATP) applications using Python. Engineered specialized signal monitoring and graphical data visualization tools directly deployed for JF-17 Thunder maintenance modules.
- **Advanced RF & Spectrum Monitoring:** Architected real-time Software-Defined Radio (SDR) interfaces for military protocol visualization. Engineered complex spectrum monitoring pipelines utilizing USRP X310 and GNU

- Radio, working hands-on with high-frequency RF systems handling up to 5 GHz bandwidth and 580 Mbps data rates.
- **Industrial Data Acquisition Systems:** Designed and implemented precise, high-fidelity sensor data acquisition architectures by programming NI myRIO and NI myDAQ hardware platforms within the LabVIEW environment.
 - **Embedded Controller & Network Integration:** Executed the integration of industrial PLC systems by developing custom APIs for the Opta Embedded Controller. Enabled robust, real-time TCP/UDP communication networks interfacing directly with LabVIEW-based control panels.
 - **Multidisciplinary R&D Problem Solving:** Demonstrated exceptional analytical abilities and technical competence by rapidly executing complex multidisciplinary tasks independently, ensuring strict attention to detail in a highly regulated defense engineering environment.

● PROJECTS

01/02/2024 – CURRENT

EKKO Device (Portable & Clinical)

- Maintained and optimized firmware for a medical device with **100% compliance** to **EN 60601-1** standards, ensuring safe clinical and portable use.
- Managed the hardware lifecycle for production-grade systems, reducing manufacturing defects by **15%** through improved **Multi-Layer PCB design** and **Signal Integrity** validation

02/02/2025 – CURRENT

Epilepsy Detection Device (EPIC)

- Developed and implemented a real-time epilepsy detection device utilizing **Machine Learning algorithms**, enhancing predictive accuracy by 30%.
- Integrated low-latency alert systems that reduced early-warning notification time by **25%**, facilitating faster medical intervention for patients

01/03/2025 – CURRENT

Underground Vehicle Profiling System using ML (FYP)

- Developed an underground vehicle classification system achieving **95% accuracy** in real-time profiling by implementing **sensor fusion** and **Edge AI** algorithms on localized hardware.
- Optimized data processing speed by **40%** by architecting a dual-version firmware (V1 & V2) that supports high-speed **SD card logging** and **edge data processing**.

01/03/2024 – 01/09/2024

Multi-Gas AQI Monitor

- Designed a comprehensive monitoring system tracking **9 different atmospheric variables** (CO2, SO2, PM2.5, etc.), ensuring **real-time data logging** with a sensor calibration error margin of less than **5%**

01/06/2025 – 01/08/2025

Python System Health Monitor

Ubuntu-based system health monitoring application with military standard ports. Features real-time temperature and performance plots per CPU core, GPS data, and Ethernet monitoring with comprehensive frontend visualization.

01/06/2025 – 01/08/2025

Arduino OPTA PLC Control

Arduino OPTA PLC programmed in Structured Text and Ladder Logic with peripheral control via TCP and UDP protocols. LabVIEW VI interface for comprehensive PLC control and monitoring.

09/09/2024 – 01/11/2024

Ergonomic Mouse PCB

Internal circuit PCB designed for ergonomic mouse with BLE connectivity. Features integrated joysticks for control and optimized component layout for ergonomic constraints within limited space.

● SKILLS

Embedded Systems

firmware | ESP32/ESP8266 | RTOS | PIC Microcontrollers | STM32

Hardware Design

High-Speed PCB Design | Signal Integrity | System on Module (SoM) | Circuit Analysis | Power Electronics | Sensor Integration | Control Systems

Communication Protocols

Wi-Fi | Bluetooth/BLE | Wiegand Interface | RFID | GRPC | TCP/UDP | USB 2.0 | SPI | I2C | BLE

Programming

C/C++ | Python | LaTeX | MATLAB/Simulink

Tools & Software

STMCube IDE | VS Code | LabVIEW | EasyEDA | Cadence Allegro X | OrCAD | Proteus | LTSpice | KiCad/Altium | Git/GitHub | Virtuoso | JLC PCB | Cadence Virtuoso

Signal Processing

Digital Signal Processing | SDR (USRP X310) | Spectral Analysis | Filter Design | Audio Processing

Emerging Technologies

Artificial Intelligence (Machine Learning, Edge AI) | Robotics (Path Planning, PID Control)

Analog/Mixed-Signal IC Design

CMOS Analog Circuit Design | Schematic & Layout Entry | Post-layout Simulation | Parasitic Extraction

HONOURS AND AWARDS

01/04/2024

EasyEDA Campus Ambassador – EasyEDA

Official campus ambassador for EasyEDA PCB design platform, promoting PCB design education and tools among students.

Link <https://scientist444.github.io/portfolio/>

01/03/2024

AirTech 24 Roboline Competition Winner – AIR University

- First-place winner with a shield and cash prize in the prestigious robotics line at AirTech 24.
- Applied **Control Systems** and **Signal Processing** to optimize robot path-following efficiency.

Link <https://scientist444.github.io/portfolio/>

Multiple Academic Excellence Awards – NUST

Consistently recognized as the best student with multiple academic excellence awards at NUST for outstanding academic performance.

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01/09/2025

President – Electrical Student Body

- Student body president for the year 2025-26. Manage all the departmental events and co-curricular activities.
- Led a team of students to organize **VoltFest**, managing departmental events and fostering industry-academia collaboration

Link <https://www.instagram.com/ee.studentcouncil/>

CMC Mentor – CMC

Participated in the CMC mentorship program, gaining valuable industry insights and professional development guidance.

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● CERTIFICATIONS

Mathworks, 11/04/2024

MATLAB Onramp Certification

Completed MATLAB Onramp certification demonstrating proficiency in MATLAB programming and computational analysis.

Mode of learning: Online

Link <https://scientist444.github.io/portfolio/>

Coursera, 27/06/2025

Cadence Design Systems: Essential Guide

Completed comprehensive training in Cadence design tools for high-speed PCB design and signal integrity analysis.

Mode of learning: Online

Link <https://scientist444.github.io/portfolio/>

Coursera, 21/07/2025

AI For Everyone Certification

Completed comprehensive course on artificial intelligence fundamentals and applications across various industries and domains.

Mode of learning: Online

Link <https://scientist444.github.io/portfolio/>

TELEDYNE LECROY, 05/09/2025

Advanced Gigabit Channel Design

Completed advanced training in gigabit-speed channel design, covering high-speed signal integrity and channel optimization techniques.

Mode of learning: Online

Link <https://scientist444.github.io/portfolio/>

TELEDYNE LECROY, 20/07/2025

High Speed Digital Design

Specialized certification in high-speed digital circuit design, signal integrity analysis, and advanced PCB design techniques.

Mode of learning: Online

Link <https://scientist444.github.io/portfolio/>

● HOBBIES AND INTERESTS

Cricket player

I am the captain of both my college cricket team and departmental team. I represent my department in four annual sports events and the college in intramural sports competitions.

Badminton

To maintain a daily athletic routine when full cricket squads are unavailable, I regularly play badminton at the college sports complex

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	C2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user